

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year: 2023

Programme: BE

Full Marks: 100

Course: Algebra and Geometry

Pass Marks: 45

Time: 3 hrs.

Candidates are required to give their answers in their own words as far as practicable. The figure in the margin indicates full marks

Attempt all the questions.

1. a) Check consistency and solve by Gauss elimination method: [7]

$$x + y + z = 6, x - y + z = 2, 2x + y - z = 1.$$

- b) Define eigen value and vector. Find the eigen value, eigenvector and diagonalize of the matrix $A = \begin{pmatrix} 2 & 1 \\ 4 & -1 \end{pmatrix}$ [8]

2. a) Solve the linear programming problem by Simplex method. [Constructing duality] [7]

$$\text{Minimize } Z = 2x_1 + 9x_2 + x_3 \text{ subject to } x_1 + 4x_2 + 2x_3 \geq 5, \quad 3x_1 + x_2 + 2x_3 \geq 4, \quad x_1 \geq 0, x_2 \geq 0, x_3 \geq 0.$$

OR

$$\text{Using simplex method maximize } Z = 150x_1 + 300x_2 \text{ subject to: } 2x_1 + x_2 \leq 16, \quad x_1 + x_2 \leq 8 \text{ and } x_1 \leq 3.5, \quad x_1 \geq 0, x_2 \geq 0.$$

$$\text{b) Solve (Big M-Method) Maximize } z = -3x_1 + 7x_2 \text{ subject to: } 2x_1 + 3x_2 \leq 5, \quad 5x_1 + 2x_2 \geq 3, \quad x_2 \leq 1. \quad [8]$$

3. a) State D'Alembert's Ratio test, show that: [7]

$$\text{i) } \sum_n \frac{1}{n} \text{ is divergent}$$

$$\text{ii) } \sum_n \frac{1}{n^{1/2}} \text{ is convergent}$$

$$\text{iii) } \sum_n \frac{(-1)^n}{n} \text{ is conditional divergent.}$$

- b) Find the center, radius and interval convergence of the power series:

$$\sum_{n=1}^{\infty} \frac{(x)^{2n+1}}{(-4)^n}.$$

4. a) Define the eccentricity of a conic section, and derive the equation of the ellipse in its standard form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. [7]

OR

Show that the line $lx + my + n = 0$ touches the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ if $a^2l^2 - b^2m^2 = n^2$. Also find the point of contact.

- b) Show the equations $16x^2 - 24xy + 9y^2 - 104x - 172y + 44 = 0$ represents a parabola. Also find the equations of axis and vertex. [8]

5. a) Define scalar triple product and reciprocal vectors. Find the reciprocal vector of: [8]

$$2\vec{i} + 3\vec{j} - \vec{k}, \quad \vec{i} - \vec{j} - 2\vec{k}, \quad -\vec{i} + 2\vec{j} + 2\vec{k}.$$

- b) Find the distance of the point (3, -4, 5) from the plane $2x + 5y + 6z = 16$ measured along a line with direction cosines proportional to 2, 1, -2. [7]

6. a) Find the shortest distance between the lines $\frac{x+2}{-4} = \frac{y}{1} = \frac{z-7}{1}$ and $\frac{x+3}{-4} = \frac{y-6}{3} = \frac{z}{2}$. Also find the equation of shortest distance. [7]

- b) Find the equation of sphere for which circle $x^2 + y^2 + z^2 + 7y - 2z + 2 = 0, \quad 2x + 3y + 4z = 8$ is a great circle. [8]

7. Attempt all the questions. [2.5 x 4]

$$\text{a) Show that } [\vec{a} \times \vec{b} \quad \vec{b} \times \vec{c} \quad \vec{c} \times \vec{a}] = [\vec{a} \quad \vec{b} \quad \vec{c}]^2.$$

- b) Find the centre of the hyperbola:

$$5x^2 - 4y^2 + 20x + 8y = 4.$$

- c) Show that the mapping. $T: R^2 \rightarrow R^2$ define by $T(x, y) = (x, x + y)$ is linear.

- d) Plane through the OX and OY include an angle α . Show that their lines of intersection lie on the cone is $(z^2 + x^2)(y^2 + z^2) \cos^2 \alpha = x^2 y^2$.

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Attempt all the questions.

- Check the given system of linear equations is consistent or not. If consistent then solve: [8]
 $x + 2y - 3z = 9, \quad 2x - y + 2z = -8, \quad 3x - y - 4z = 3.$
 - Verify Cayley Hamilton theorem for the matrix [7]

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$
- Define basis for a vector space. Show that the vectors (1, 2, 1), (2, 1, 0), (1, -1, 2) for a basis of $\mathbb{R}^3$. [7]
 - Find Eigen value and corresponding Eigen vectors of the matrix. [8]

$$A = \begin{pmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{pmatrix}$$

OR

Define Rank of matrix. Find Rank of following matrix by using Echelon form:

$$\begin{pmatrix} 2 & 1 & 3 \\ 4 & 7 & 13 \\ 4 & -3 & -1 \end{pmatrix}$$

- Using simplex method, solve the following LPP: [7]
 Maximize $Z = 2x_1 - x_2 + 2x_3$ subject to constraints
 $x_1 + 2x_2 - 2x_3 \leq 20, \quad 2x_1 + x_2 \leq 10, \quad x_2 + 2x_3 \leq 5$ and $x_1, x_2, x_3 \geq 0$

OR

Construct the dual problem and solve by simplex method:

Minimize $Z = 8x_1 + 9x_2$ subject to constraints
 $x_1 + x_2 \geq 6, \quad 3x_1 + x_2 \geq 21$ and $x_1, x_2 \geq 0$

- State and prove Leibnitz's theorem for convergence of an infinite series. [4]
- Test the convergence and divergence of the following series by using D'Alembert ratio test: [4]

$$\frac{1}{2} + \frac{4}{2^2} + \frac{9}{2^3} + \dots$$
- Define scalar and vector product three vectors. Derive the expression for vector product of three vectors. [8]
 - Find the set of reciprocal vector of $a = 2i + 3j - k$, $b = i - j - 2k$ and $c = -i + 2j + 2k$. [7]
- Define conic section and derive the standard equation of Ellipse. [8]
 - Obtain the condition that the line $lx + my + n = 0$ may touch the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. [7]
- Show that the lines $\frac{x+4}{3} = \frac{y+6}{5} = \frac{1-z}{2}$ and $3x - 2y + z + 5 = 0 = 2x + 3y + 4z$ are coplanar. Find their point of intersection and the plane in which they lie. [7]
 - Find the centre and radius of circle $x^2 + y^2 + z^2 + 12x - 13y - 16z + 111 = 0, \quad 2x + 2y + z = 17$. [8]
- Attempt all: (2 x 5)
 - Find the equation of the sphere one of whose great circle is:
 $x^2 + y^2 + z^2 = 4, \quad x^2 + y^2 + z^2 - 2x + 4y - 6z = 11$
 - Show that the points (4, 5, 1), (3, 9, 4), (-4, 4, 4) and (0, -1, -1) are coplanar.

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Attempt all the questions.

1. a) Check the given system of linear equations is consistent or not, if consistent then solve: [7].
 $x + y + z = 1, \quad x + 2y + 3z = -4, \quad x + 3y + 7z = 13.$

- b) Find eigenvalue and eigenvector of the matrix, $\begin{bmatrix} 2 & 1 & 2 \\ 0 & -1 & 3 \\ 0 & 1 & 1 \end{bmatrix}$. [8]

2. a) i. Let $V = \mathbb{R}^2$ be a vector space, show that $W = \{(x, y) : x + 2y = 0\}$ is a vector sub-space of V . [3]
 ii. Check the vectors $(1, 2, 1), (2, 1, 0), (1, -1, 2)$ for a basis for $\mathbb{R}^3$ or not. [4]

- b) Solve the given LPP by using appropriate simplex method. [8]

$$\text{Max } Z = 2x_1 + x_2$$

$$\text{S.t. } 2x_1 - x_2 \geq 2; \quad x_1 - x_2 \leq 2; \quad x_1 + x_2 \leq 4; \quad x_1 \geq 0, x_2 \geq 0.$$

OR

Construct the dual problem and solve by simplex method.

$$\text{Minimize } Z = 8x_1 + 9x_2 \text{ subject to constraints } x_1 + x_2 \geq 6, \quad 3x_1 + x_2 \geq 21, \quad x_1, x_2 \geq 0.$$

3. a) State and prove p-series test. [7]
 b) Find the interval and radius of convergence of the series, [8]

$$\sum_{n=0}^{\infty} \frac{(x-2)^n}{10^n}.$$

4. a) Find the value of λ when the line $y = 2x + \lambda$ is tangent to the hyperbola $3x^2 - y^2 = 3$. [7]
 b) Show that the equations $9x^2 - 24xy + 16y^2 - 50x - 100y + 225 = 0$ represents a parabola. Also, determine its equation of axis, vertex, latus rectum, focus and directrix. [8]

5. a) Examine, whether the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-4}{4}$ and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ are coplanar or not. If so, find the equation of plane containing them. [7]

OR

Find the length and equation of the shortest distance between the lines

$$\frac{x}{2} = \frac{y}{-3} = \frac{z}{1} \quad \text{and} \quad \frac{x-2}{3} = \frac{1-y}{5} = \frac{z+2}{2}.$$

- b) Find the centre and radius of circle $x^2 + y^2 + z^2 + 12x - 13y - 16z + 111 = 0, \quad 2x + 2y + z = 17$. [8]

6. a) Find the equation of right circular cylinder whose guiding curve is $x^2 + y^2 + z^2 - x - y - z = 0, \quad x + y + z = 1$. [7]

- b) i. If a, b, c be any three vectors. Show that $[a+b \quad b+c \quad c+a] = 2[a \quad b \quad c]$. [4]

- ii. Define reciprocal system of vectors. Find the reciprocal of a for $a = (-1, 2, 2), b = (2, 3, 1)$ and $c = (1, -1, -2)$. [4]

7. Attempt all the questions. [2.5 x 4]

- a. Find the area of the triangle whose vertices are $(-1, 1, 2), (2, 5, -1), (1, 2, 3)$.
 b. Define cross product of four vectors.
 c. Find the equation of plane through the origin and having direction cosines proportional to $(5, 2, -3)$ and $(2, 1, -2)$
 d. For what value of 'k' the plane $x - 2y - 2z = k$ touches the sphere $x^2 + y^2 + z^2 + 4y - 2x - 6z = -5$.

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Attempt all the questions.

1. a) Check whether the given system of linear equations is consistent or not, if consistent then solve the system of linear equations: [7]
 $2x + 5y + 6z = 13, \quad 3x + y - 6z = 13, \quad x - 3y - 8z = -13.$
 b) [4+4]
 i) State Cayley Hamilton theorem. Verify it for the matrix $A = \begin{pmatrix} 1 & -2 \\ 4 & 5 \end{pmatrix}$.
 ii) Check whether the set of vectors $\{(1,1,0), (1,0,1), (3,1,1)\}$ for a basis for $\mathbb{R}^3$ or not.
2. a) Define eigen value and eigen vector. Find the eigen values and corresponding eigen vectors of $\begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix}$. [7]
 b) Convert the following primal LPP into dual LPP and solve by using simplex method: [8]
 Minimize $Z = 8x_1 + 9x_2$ subject to $x_1 + 3x_2 \geq 4, \quad 2x_1 + x_2 \geq 5, \quad x_1 \geq 0, x_2 \geq 0.$
3. a) Find the set of reciprocal system of vectors to $a = 2i + 3j - 2k, b = i - j - 2k$ and $c = -i + 2j + 2k$. [7]
 b) Find the interval of convergence, centre of convergence and radius of convergence of an infinite series $\sum \frac{n^2}{2^n} (x + 4)^n$. [8]
4. a) State Cauchy root test. Test the convergence of the following infinite series: [7]
 i. $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots$

ii. $\sum \left(\frac{2n}{n+1} \right)^n$

- b) Find the vertex, eccentricity, foci, and equation of directrix of the ellipse: $3x^2 + 4y^2 - 12x - 8y + 4 = 0$. [8]
5. a) Find the condition for the line $y = mx + c$ to be the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Also, find the point of contact. [7]

OR

Sketch and describe the polar conic $r = \frac{12}{3+2 \cos \theta}$.

- b) Define skew lines. Find the shortest distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$. Also, find the equation of the line of shortest distance. [8]
6. a)
 i. Find the equation of the cone with vertex at the origin and which passes through the curve of intersection of $ax^2 + by^2 + cz^2 = 1$ and $lx + my + nz = p$. [2]
 ii. Find the equation of the right circular cylinder of radius 2 whose axis is the line $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-3}{2}$. [5]
 b) Find the centre and radius of the circle $x^2 + y^2 + z^2 - 8x + 4y + 8z - 45 = 0, \quad x - 2y + 3z = 3$. [8]
- OR**
- Show that the plane $2x - 2y + z = -12$ touches the sphere $x^2 + y^2 + z^2 - 2x - 4y + 2z = 3$. Also find the point of contact.
7. Attempt all the questions: [2.5 x 4]
 a. Check whether the transform $T: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $T(x, y) = x + y$ is linear or not.
 b. Find the volume of the parallelepiped whose concurrent edges are given by: $\vec{a} = \vec{i} + 2\vec{j} + 3\vec{k}, \vec{b} = 3\vec{i} + 4\vec{j} - 5\vec{k}, \vec{c} = \vec{i} - 2\vec{j} + 3\vec{k}$.
 c. Transform the equation $x^2 + 3xy + y^2 = 0$ in which the origin is transformed to (2, 3) with axes remaining parallel to the old axes.
 d. Show that the line joining the points (-2, 1, 3) and (1, -3, 4) is parallel to the plane $2x + 3y + 6z + 5 = 0$.